Reviewer Pack — Minimal Rank of K-theory over Function Fields vs Resolution …

Entry e249e7fc786f · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

2026-05-25 03:40:03 UTC

Minimal Rank of K-theory over Function Fields vs Resolution Proof Length for Random k-CNF

Entry ID: `e249e7fc786f` **Verdict**: `INCONCLUSIVE` **Recorded**: 2026-05-25 03:40:03 UTC

1. Conjecture

Field A (mathematical branch): Algebraic Geometry (K-theory) **Field B** (complexity object): Complexity Theory: Resolution Proof Complexity

Statement:

> ['For a random k-CNF instance with n variables, the minimal rank of the K-theory over its function field is asymptotically equal to 2^Ω(n).', 'Equivalently, for all instances with property P, the resolution proof length is at least 2^(n - ω(n)) where ω(n) is the slowest growing function such that 2^ω(n) ≤ n.', 'A single counterexample instance with minimal K-theory rank less than 2^(n - ω(n)) or resolution proof length less than 2^(n - ω(n)) refutes this conjecture.']

Rationale (proposer's reasoning):

> ['K-theory is a well-established branch of algebraic geometry that studies vector bundles and other high-dimensional objects. Its minimal rank can reflect the complexity of the underlying structure.', 'Since K-theory is closely related to the classification of vector bundles, it may capture aspects of computational complexity not evident in lower-level algebraic structures.', 'The proposed conjecture would connect a deep geometric concept with resolution proof length, potentially providing new insights into the complexity of satisfiability problems.']

Taxonomy category: `TSEITIN_RES` (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): `eb2b2fb93c19b33e`

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

> For a given k-CNF instance, if the minimal rank of K-theory over its function field is less than 2^(n - ω(n)) or the resolution proof length is less than 2^(n - ω(n)), the conjecture is falsified.

3. Barrier filter (F1)

Final: `PASS`

Barrier	Verdict	Confidence	LLM_1	LLM_2
RELATIVIZATION	UNCERTAIN	1.00	HITS	UNCERTAIN
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	UNCERTAIN	1.00	HITS	UNCERTAIN
KARP_LIPTON	SAFE	0.50	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: `ADJACENT_OK` against 10 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - `'K-theory over function fields' AND 'resolution proof complexity'` - `'algebraic geometry K-theory' AND 'k-CNF resolution proof length'` - `'asymptotic minimal rank K-theory' AND 'random k-CNF resolution complexity'`

Top relevant hits considered: - [http://arxiv.org/abs/1311.1421v3] Multiplicative differential algebraic K-theory and applications - [http://arxiv.org/abs/1502.02240v2] On the K-theory of linear groups - [http://arxiv.org/abs/1811.09564v3] Dévissage for Waldhausen K-theory - [http://arxiv.org/abs/1510.06118v3] G-theory of root stacks and equivariant K-theory - [http://arxiv.org/abs/1208.1599v1] Algebraic K-theory of endomorphism rings - [http://arxiv.org/abs/2001.01608 The plethory of operations in complex topological K-theory - [http://arxiv.org/abs/2104.12736v2] Deformation theory of perfect complexes and traces - [http://arxiv.org/abs/2311.02318v2] The connecting homomorphism for Hermitian K-theory

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)
Sandbox: isolated subprocess, stdlib-only, ≤90s wall time per cycle **Execution**: rc=1, elapsed=0.1s

5.1 Generated Python source

```python
import random
import math
from fractions import Fraction

def gaussian_elimination(matrix):
    n = len(matrix)
    for i in range(n):
        # Find pivot row
        max_row = i
        for j in range(i+1, n):
            if abs(matrix[j][i]) > abs(matrix[max_row][i]):
                max_row = j
        matrix[i], matrix[max_row] = matrix[max_row], matrix[i]

        # Eliminate below the pivot
        for j in range(i+1, n):
            factor = Fraction(matrix[j][i], matrix[i][i])
            for k in range(n):
                if k < i:
                    matrix[j][k] -= factor * matrix[i][k]
                elif k == i:
```

```
                    matrix[j][k] = 0
                else:
                    matrix[j][k] -= factor * matrix[i][k]
    return matrix

def resolution_length(cnf_instance):
    n = len(cnf_instance)
    clauses = cnf_instance[:]
    stack = []

    while True:
        if not clauses:
            return len(stack)

        unit_clause = next((c for c in clauses if len(c) == 1), None)
        if unit_clause:
            literal = unit_clause[0]
            stack.append(literal)
            clauses = [c for c in clauses if literal not in c and -literal not in c]
        else:
            resolvent = []
            for i in range(len(clauses)):
                for j in range(i+1, len(clauses)):
                    common_literals = set(c for c in clauses[i] if -c in clauses[j])
                    if common_literals:
                        resolvent = [l for l in clauses[i] if l not in common_literals]
                        resolvent.extend(l for l in clauses[j] if l not in common_literals)
                        break
                else:
                    continue
                break
            if not resolvent:
                return len(stack)
            clauses.append(resolvent)

    return len(stack)

def run_trial(seed: int) -> dict:
    random.seed(seed)
    n = random.choice([5, 10, 15, 20, 30, 40])
    cnf_instance = [[random.randint(-n, n) for _ in range(random.randint(2, n))] for _ in range(n)

    rank = gaussian_elimination(cnf_instance)
    resolution_len = resolution_length(cnf_instance)

    metric_value = max(len([x for x in row if x != 0]) for row in rank)
    conjecture_holds = (metric_value >= 2**(n - math.log(n, 2))) and (resolution_len >= 2**(n - ma
    counterexample = "" if conjecture_holds else "K-theory rank or resolution length too small"

    return {
        "metric_name": "Minimal Rank of K-theory vs Resolution Proof Length",
        "metric_value": metric_value,
        "instances_tested": 1,
        "conjecture_holds": conjecture_holds,
        "counterexample": counterexample
    }
```

```
if __name__ == "__main__":
    import sys
    seeds = [int(s) for s in sys.argv[1:]] if sys.argv[1:] else [2**i + 1 for i in range(5, 8)]

    results = []
    for seed in seeds:
        result = run_trial(seed)
        print(f"TRIAL: {result}")
        results.append(result)

    mean_value = sum(r["metric_value"] for r in results) / len(results)
    std_value = math.sqrt(sum((r["metric_value"] - mean_value)**2 for r in results) / len(results)
    support_fraction = sum(1 for r in results if r["conjecture_holds"]) / len(results)

    if all(r["conjecture_holds"] for r in results):
        print(f"RESULT: SUPPORTED mean={mean_value} std={std_value} support_fraction={support_frac
    elif support_fraction >= 0.8:
        print(f"RESULT: SUPPORTED mean={mean_value} std={std_value} support_fraction={support_frac
    else:
        first_failing_seed = next((r["seed"] for r in results if not r["conjecture_holds"]), None)
        print(f"RESULT: FALSIFIED counterexample=\"K-theory rank or resolution length too small\"
```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_91ffa478.py", line 98, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_91ffa478.py", line 77, in run_trial
    rank = gaussian_elimination(cnf_instance)
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_91ffa478.py", line 37, in gaussian_elim
    matrix[j][k] -= factor * matrix[i][k]
                             ~~~~~~~~~^^^
IndexError: list index out of range
```

8. Critic adversarial review

Critic verdict: `CHALLENGE`

Critic reasoning:

> skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: `INCONCLUSIVE`

Reasoning:

The test crashed before producing data, which means that it was unable to complete its execution and provide a result. | next: Re-run the test with appropriate error handling to ensure it completes without crashing.

11. Audit log (LLM calls)

Total LLM calls: 9

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	14312
2	preregistration	ollama_remote	glm4:latest	0	0	6065
3	novelty	ollama_remote	glm4:latest	0	0	4769
4	novelty	ollama_remote	glm4:latest	0	0	6176
5	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	28071
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	12089
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	10517
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	11430
9	judge	ollama_remote	glm4:latest	0	0	8303

Totals: 0 input tokens, 0 output tokens, ~$0.0000 cost, 101730 ms total latency. Provider mix: {'ollama_remote': 9}

(full prompt+response transcripts available in `research/audit/e249e7fc786f.jsonl`)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/e249e7fc786f.jsonl` for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: `research/pvsnp_sandbox/test_*.py` (per-cycle)
Replay tarball: `research/replay/e249e7fc786f.tar.gz` (if generated)